Problem-Solution Fit Canvas

Date	1 July 2026
Team ID	—
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas for Rising Waters, mapping the target audience's limitations, pains and behaviors against the proposed solution.

1. Customer Segment(s) [CS]	6. Customer Limitations [CL]	5. Available Solutions [AS]
Residents of flood-prone river-basin communities; local disaster-management officers	Limited access to real-time hydrological data; low digital literacy for complex dashboards; unreliable internet in rural areas	Government flood bulletins (generic, delayed); word-of-mouth; manual river-gauge checks — none combine multiple factors into one risk score

2. Problems / Pains + Frequency [PR]	9. Problem Root / Cause [RC]	7. Behavior + Intensity [BE]
Flood warnings arrive too late or are too generic; happens every monsoon season (seasonal, high frequency)	No accessible tool combines rainfall, river level, drainage and terrain into a single, immediate risk assessment	Residents rely on visual river checks and neighbourhood chatter; high intensity of concern during heavy rain spells

3. Triggers to Act [TR]	10. Your Solution [SL]	8. Channels of Behavior [CH]
Sudden heavy rainfall, rising river levels, official weather alerts	A Flask web app where users enter current weather/terrain readings and instantly receive a Flood / No-Flood prediction with probability, powered by a trained ML model (XGBoost, ~75% test accuracy)	Online: web browser on phone/desktop Offline: shared verbally within the community after checking the app

4. Emotions Before / After [EM]
Before: anxious, uncertain, reliant on guesswork. After: informed and prepared — able to make an evacuation or safety decision backed by a concrete risk assessment.